



PVT 掌锐电子



SF DEBUG方式

1、Disable HW overlays or service call SurfaceFlinger 1008 i32 1 可以全推GPU 合成以判断问题是否与合成方式有关。（投屏录屏截屏也可用于辅助验证）

2、sfdump是一个执行简单，且内容很丰富的常规debug信息dump手段，包含了当前合成的layer信息，BufferQueue的信息，合成的信息等等

```
adb shell dumpsys SurfaceFlinger > dumpsf.log
```

3、抓取winscope和perfetto分析，主要用于分析卡顿卡死或者闪屏问题。

4、打开BQ_LOG日志

SF DEBUG方式

5、BQDump

Dump图层画面

先执行命令

```
adb root
```

```
adb remount
```

```
adb push libgralloc_extra_sys.so /system_ext/lib64/
```

```
adb shell sync
```

```
adb shell setprop debug.mediatek.disp_decompress 0
```

```
adb shell stop
```

```
adb shell start
```

复现到问题后抓取bq数据

dump1.bat 是抓一帧的数据

dump10.bat 能够抓10帧的数据

SF DEBUG方式

6、SF内存占用异常

```
adb shell " mv vendor/lib64/hw/mapper.minigbm_mtk.so vendor/lib64/hw/mapper.minigbm_mtk.so.bak"
```

```
adb shell sync
```

```
adb reboot
```

复现问题

```
adb shell dumpsys SurfaceFlinger > dumsf.log
```

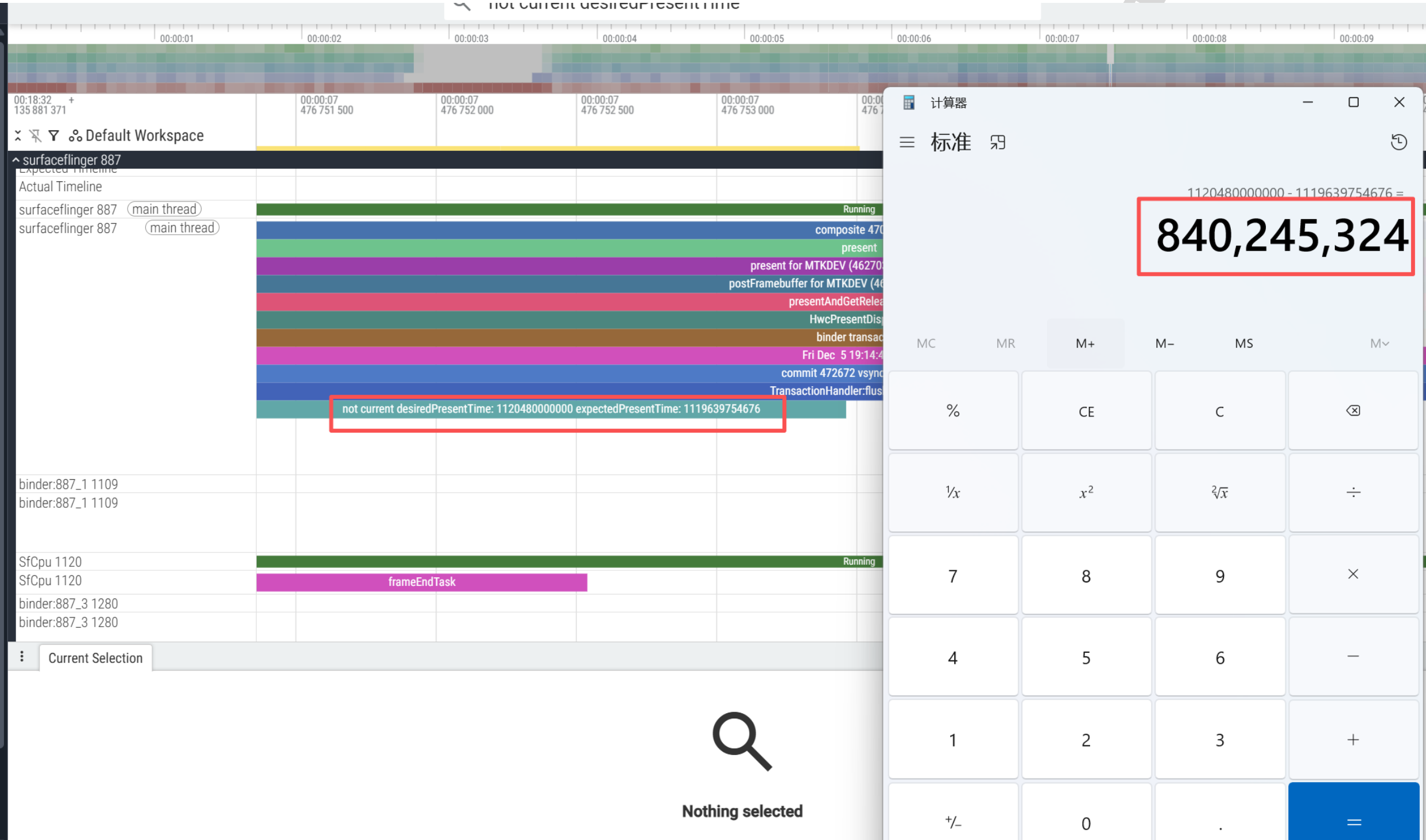
```
adb shell dumpsys meminfo surfaceflinger > memsf.txt
```

SF DEBUG方式



SF DEBUG案例

问题：爱奇艺长按2倍速播放数分后，视频卡顿，画面冻结，时间进度正常增加
设置了desiredPresentTime且大于expectedPresentTime

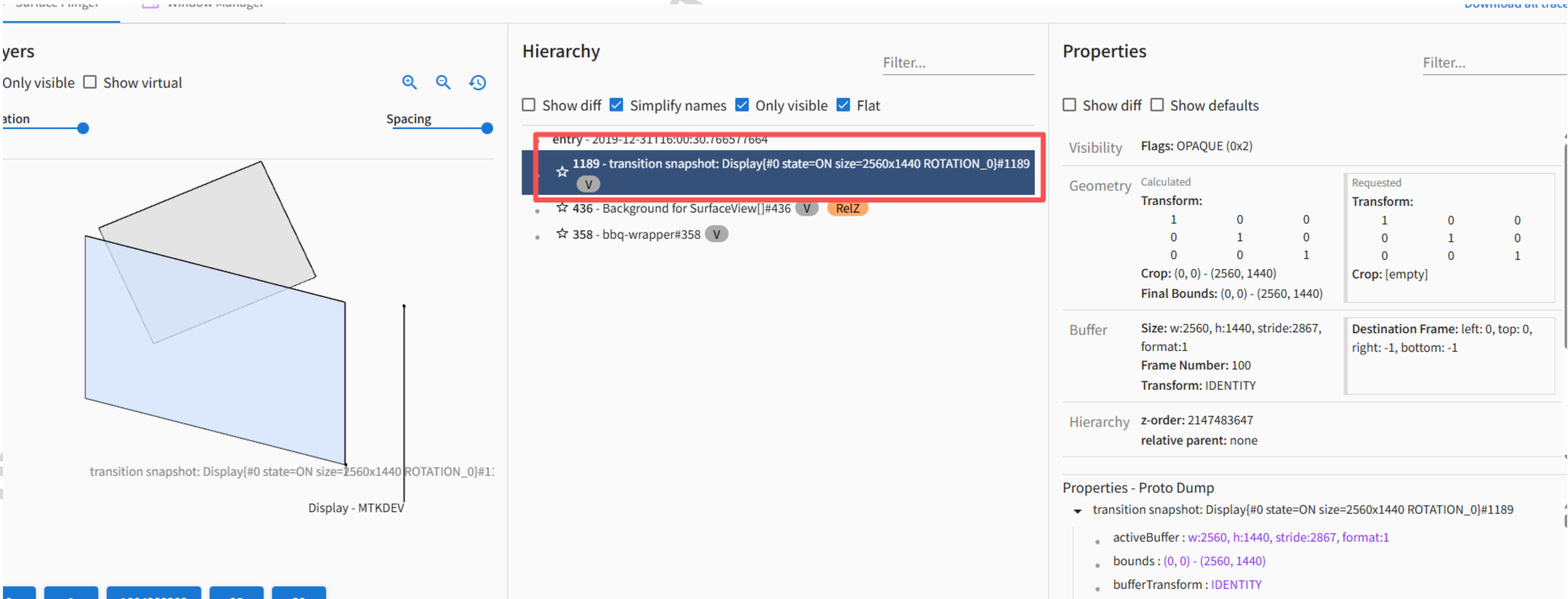


```
TransactionHandler::TransactionReadiness SurfaceFlinger::transactionReadyTimelineCheck(  
    const TransactionHandler::TransactionFlushState& flushState) {  
    using TransactionReadiness = TransactionHandler::TransactionReadiness;  
    const auto& transaction = *flushState.transaction;  
    TimePoint desiredPresentTime = TimePoint::fromNs(transaction.desiredPresentTime);  
    // Do not present if the desiredPresentTime has not passed unless it is more than  
    // one second in the future. We ignore timestamps more than 1 second in the future  
    // for stability reasons.  
    if (!transaction.isAutoTimestamp && desiredPresentTime >= mExpectedPresentTime &&  
        desiredPresentTime < mExpectedPresentTime + 1s) {  
#ifdef MTK_SF_DEBUG_SUPPORT  
        if (flushState.bTransactionIsReadyToBeAppliedLog) {  
            ALOGD("%s", String8::format("not current d=%" PRId64 ", e=%" PRId64  
                , desiredPresentTime.ns(), mExpectedPresentTime.ns()).string());  
        }  
#endif  
        ATRACE_FORMAT("not current desiredPresentTime: %" PRId64 " expectedPresentTime: %" PRId64,  
            , desiredPresentTime, mExpectedPresentTime);  
        return TransactionReadiness::NotReady;  
    }  
}
```

SF DEBUG案例

问题：DVR画面语音切换系统语言，DVR预览画面卡顿

横竖屏切换、系统语言切换、日夜模式切换。system_server会截个屏，然后创建一个层级很高的图层显示这个截图，这样一来切换操作时，用户看到的就是截图的内容，目的是为了防止UI 混乱。，切换完成后，移除截图图层，可以看到正常画面。



SF DEBUG案例

问题：进入STR后唤醒，中控比仪表屏延迟23s左右点亮

Layer创建超过4096， 从日志中看到大量Activity启动时创建的Task都未销毁

```
4415 E SurfaceFlinger: AddClientLayer failed, mNumLayers (4098) >= MAX_LAYERS (4096), mLayerTracker (4098), mBufferCountTracker (167), mMirrorDisplayHandleTracker(0), mLayersPending
4415 E SurfaceFlinger: AddClientLayer already dumped sampling 34(ms) before
6990 E SurfaceComposerClient: SurfaceComposerClient::createSurface error Out of memory
```

左边为创建数量，右边为销毁数量，该问题是打开persist.wm.debug.shell_transit导致，最终是应用端修改

[illegible]

SF DEBUG案例

问题：硬按键重启过程中system_server发生crash
app执行nativeGetLatestVsyncEventData时因surfaceflinger退出返回NULL报的空指针。

```
[ 254.681805][T500001] init: init 6: [254524][12]Sending signal 15 to service 'surfaceflinger'
(pid 915) process group...
```

因此无法获取vsync相关信息

```
10-01 07:42:31.038 3190 5140 W SurfaceComposerClient: ComposerService aidl remote
(surfaceflinger) died [0xb400007350ca20e0]
```

```
10-01 07:42:31.040 1604 1668 E DisplayEventReceiver: Failed to get latest vsync event data:
```

```
10-01 07:42:31.040 1604 1668 W DisplayEventReceiver: Failed to get latest vsync event data
from surface flinger
```

这里直接返回null，因此报错

SF DEBUG案例

问题：执行长时间压力测试后，surfaceflinger存在疑似内存泄漏
按照SF内存占用异常 dump信息发现是图层和图层buffer未释放，该问题是多屏移屏动画导致

```
planes: R/G/B/A:      w/h:2560x1440, stride:10240 bytes, size:14976000
+ name:VRI[AppWidgetWindowLayout]#206(BLAST Consumer)206, id:106041, size:14626.07KiB, w/h:2560x1440, usage: 0xb00, req fmt:1, fourcc/mod:875708993/576460752303423602, dataspace: 0x8810000, compressed: true
  planes: R/G/B/A:      w/h:2560x1440, stride:10240 bytes, size:14976000
+ name:FramebufferSurface, id:13, size:2744.07KiB, w/h:1440x480, usage: 0x1b00, req fmt:1, fourcc/mod:875708993/576460752303423602, dataspace: 0x0, compressed: true
  planes: R/G/B/A:      w/h:1440x480, stride:5760 bytes, size:2808832
+ name:VRI[AppWidgetWindowTouchProxyLayout]#205(BLAST Consumer)205, id:106037, size:4226.07KiB, w/h:2560x416, usage: 0xb00, req fmt:1, fourcc/mod:875708993/576460752303423602, dataspace: 0x8810000, compressed: true
  planes: R/G/B/A:      w/h:2560x416, stride:10240 bytes, size:4326400
+ name:VRI[CeilingLauncher]#139(BLAST Consumer)139, id:104224, size:14626.07KiB, w/h:2560x1440, usage: 0xb00, req fmt:1, fourcc/mod:875708993/576460752303423602, dataspace: 0x8810000, compressed: true
  planes: R/G/B/A:      w/h:2560x1440, stride:10240 bytes, size:14976000
+ name:VRI[AppWidgetWindowTouchProxyLayout]#170(BLAST Consumer)170, id:105003, size:4226.07KiB, w/h:2560x416, usage: 0xb00, req fmt:1, fourcc/mod:875708993/576460752303423602, dataspace: 0x8810000, compressed: true
  planes: R/G/B/A:      w/h:2560x416, stride:10240 bytes, size:4326400
Total imported by gralloc: 4399321.29KiB
FlagManager values:
demo flag: -1
```

PVT

Thank You

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